TDR Hacking Resistive Terminations - Lab 2.3

ECEN 5534 · Signal Integrity Measurements

Module 2 · Fall 2026

A VNA-based virtual TDR sets what we can and cannot see

NanoVNA-F V3 specifications

Specification	Value
Instrument	NanoVNA-F V3 portable VNA
Frequency range	1 MHz – 6 GHz
Dynamic range	S21: 65 dB, S11: 50 dB
Ports / connector	2 ports (S11, S21), SMA
Calibration	1-port SOL at the SMA reference plane
Measured	S11 of Open, SMD 0 mil, SMD 400 mil, TH 400 mil

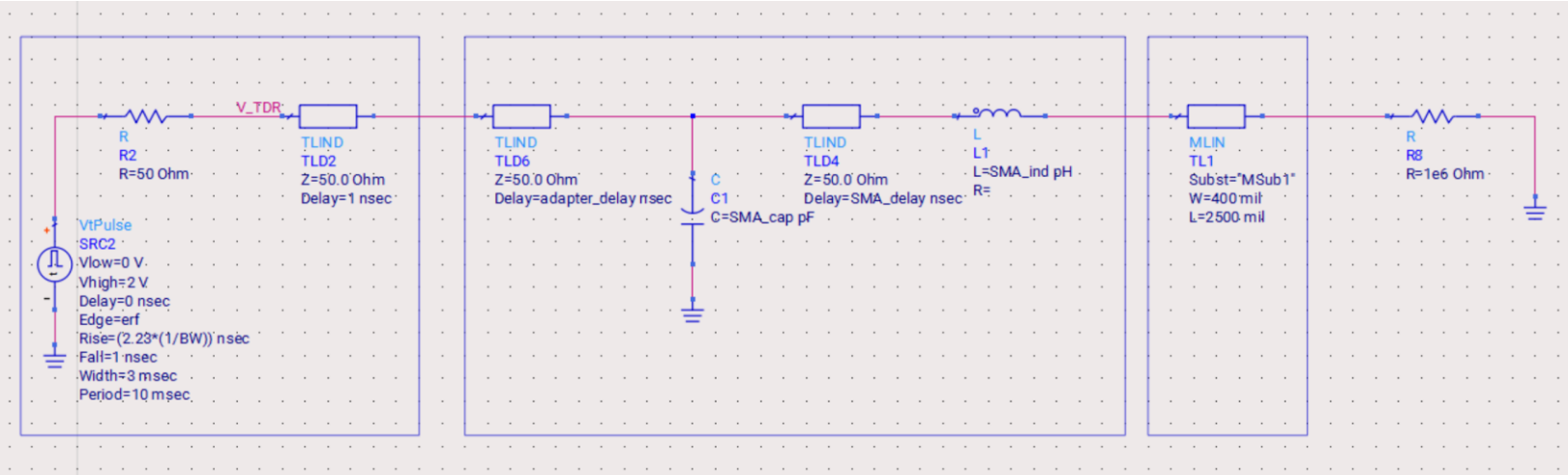
Virtual TDR settings (ADS)

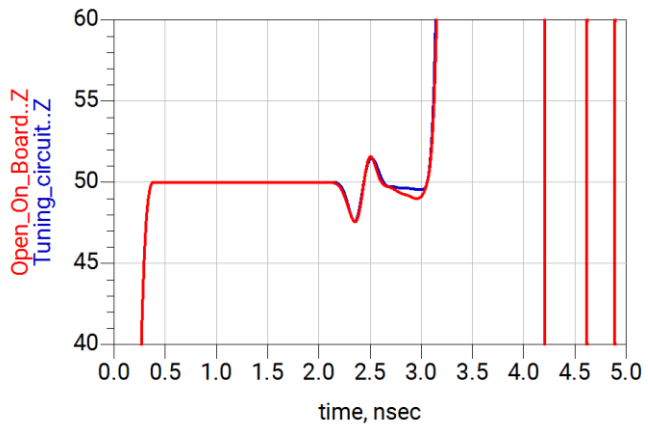
- Bandwidth 6 GHz → 10–90 rise time = 1/BW ≈ 167 ps
- ADS erf edge is 0–100: Rise = 2.23/BW ≈ 372 ps
- Step source: 0 → 2 V behind 50 Ω, 1 ns lead-in line
- $Z(t) = 50 \cdot V/(2 - V)$

SO, WHAT

A 167 ps edge resolves features about 540 mil apart on microstrip; anything under about 180 mil acts lumped.

The Open standard calibrates the fixture model



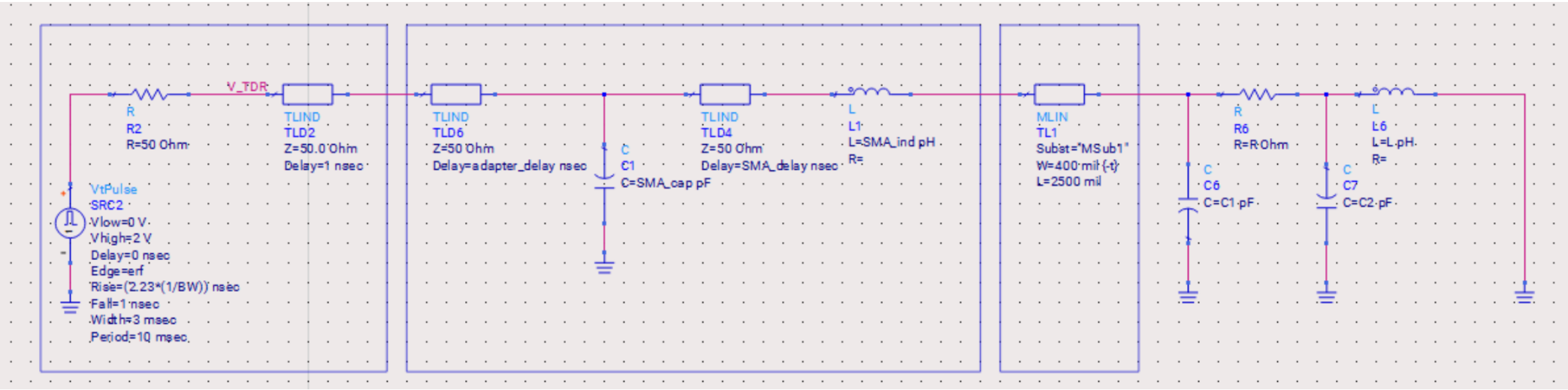


Fixture parameter	Value
Adapter delay	0.085 ns
SMA cap	0.188 pF
SMA delay	0.0475 ns
SMA ind	269 pH
Line (MLIN)	W 400, L 2500 mil

SO, WHAT

Model tracks the open within about 0.5 Ω through the SMA dip (47.5 Ω) and bump (51.5 Ω). These values are the starting point for every resistor case.

One resistor model for every case: C1 – R – C2 – L



Resistor network

- C1: pad capacitance at the launch
- R: DC value of the resistor
- C2 + L: body and loop to ground

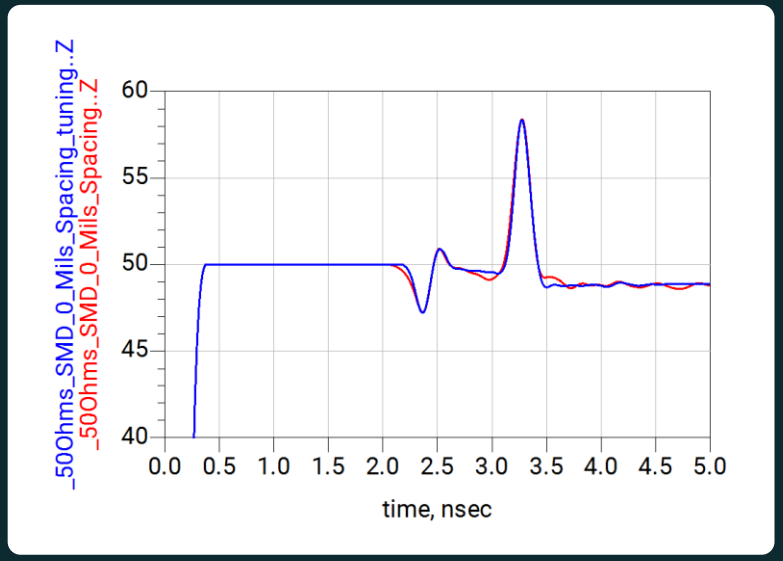
Frequency-domain comparison

- Z_meas = 50·(1+S11_meas)/(1−S11_meas)
- Z_sim = 50·(1+S11_sim)/(1−S11_sim)
- Similarity = |Z_meas − Z_sim|

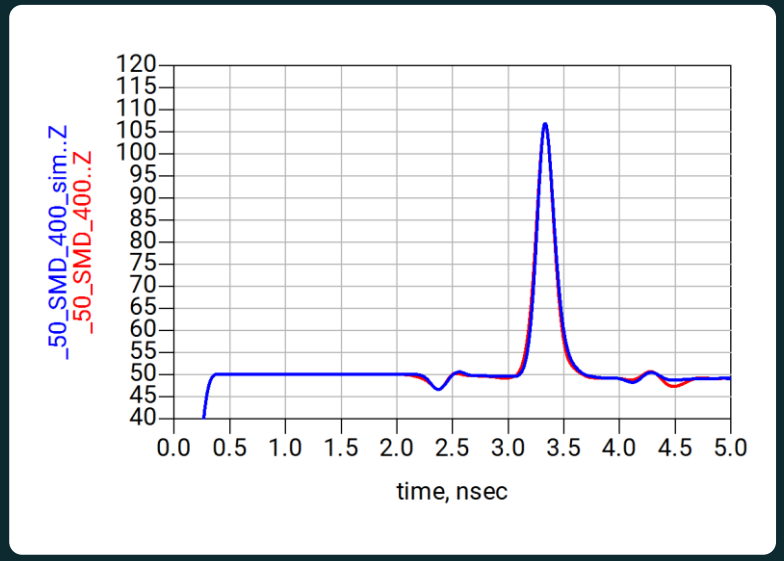
SO, WHAT

Only C1, R, C2 and L change between cases, so every difference in the plots comes from the resistor and its loop.

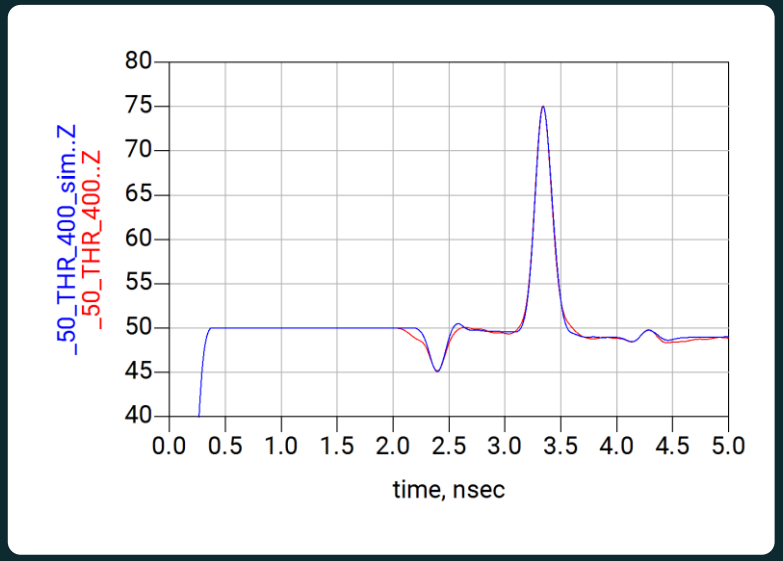
The TDR shows every resistor as an inductive bump



SMD, 0 mil



SMD, 400 mil



Through-hole, 400 mil

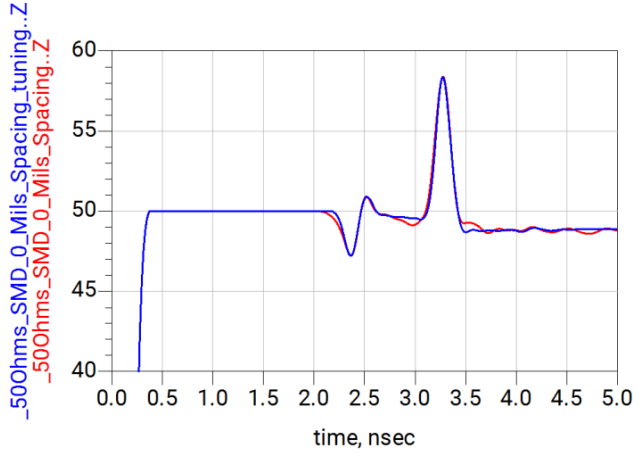
Measured peaks

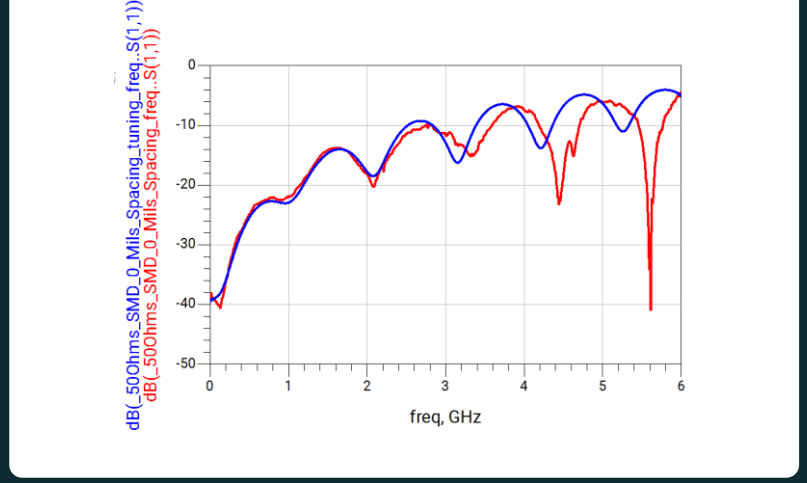
- SMD 0 mil: about 58 Ω, settles near 48.8 Ω
- SMD 400 mil: about 107 Ω
- TH 400 mil: about 75 Ω (deeper launch dip, 45 Ω)

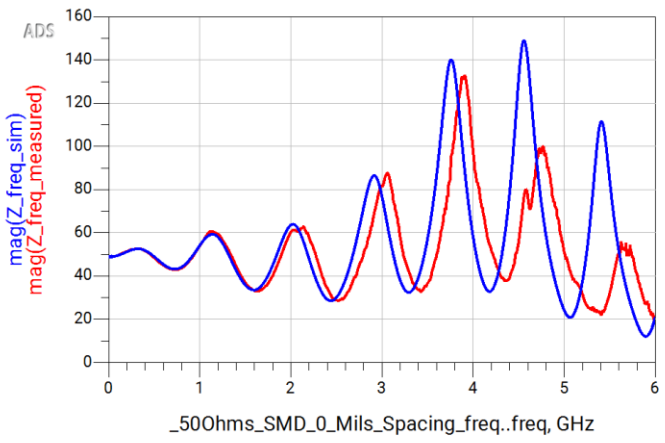
SO, WHAT

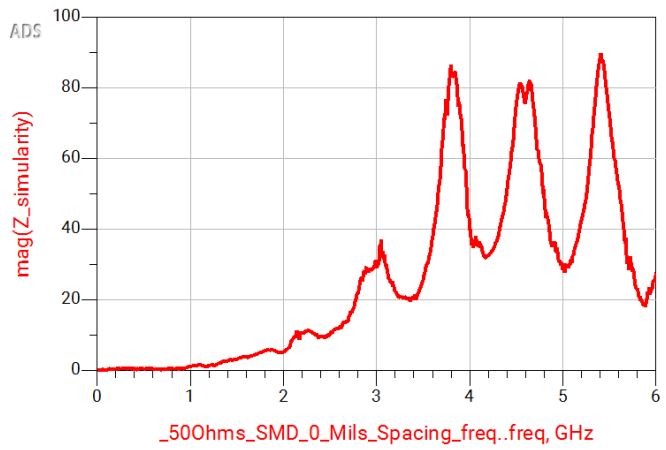
Same DC resistance, very different fast-edge behavior: the 400 mil path to ground almost doubles the peak, so loop geometry sets the inductance.

SMD 50 Ω, 0 mil: model holds up to about 2 GHz









Element	Fitted value
R	48.95 Ω
L	2.59 nH
C1 / C2	0.39 / 0.01 pF
SMA cap / ind	0.198 pF / 150 pH
Adapter / SMA delay	0.090 / 0.050 ns

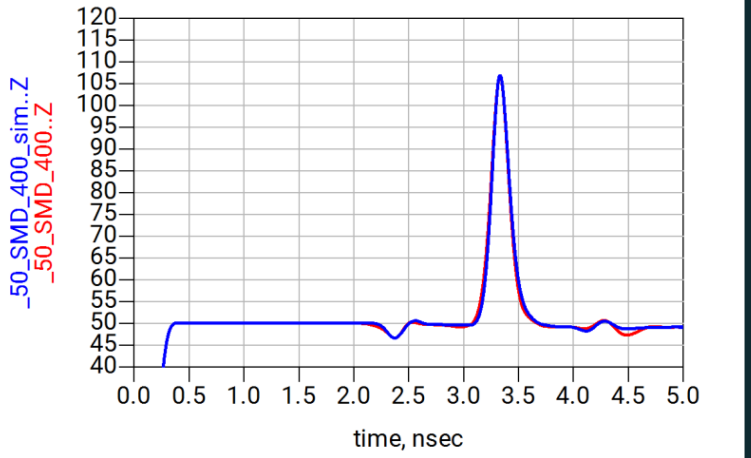
Key results

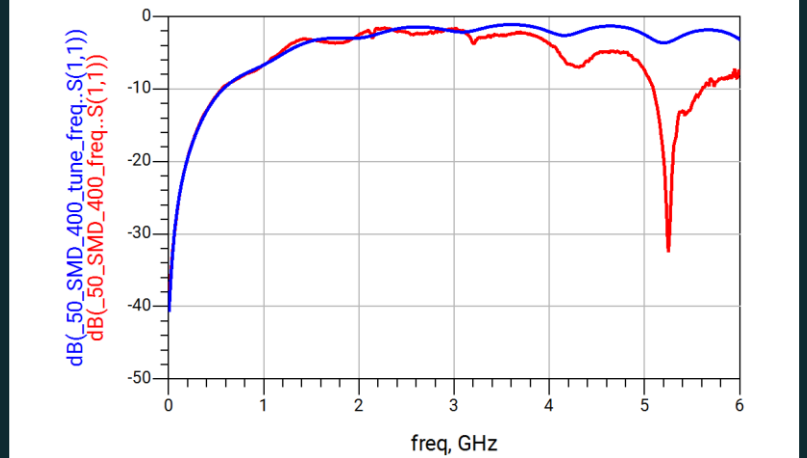
- TDR peak about 58 Ω, model on top of it
- |S11| within about 1 dB to 2 GHz
- |ΔZ| under about 5 Ω to 2 GHz

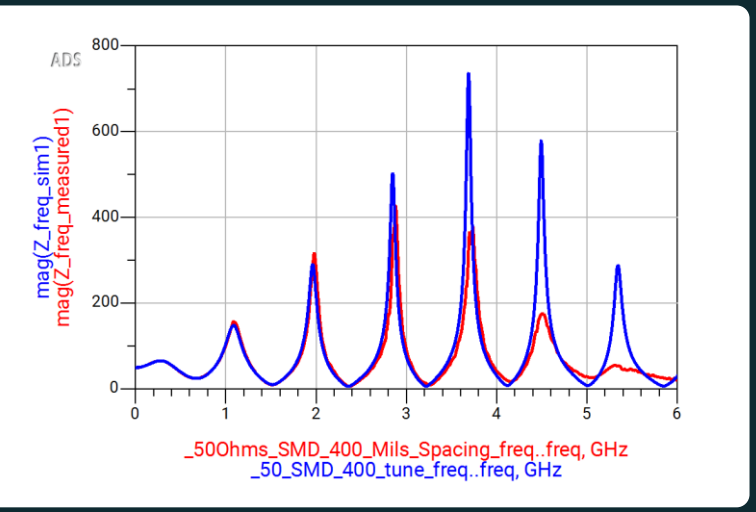
SO, WHAT

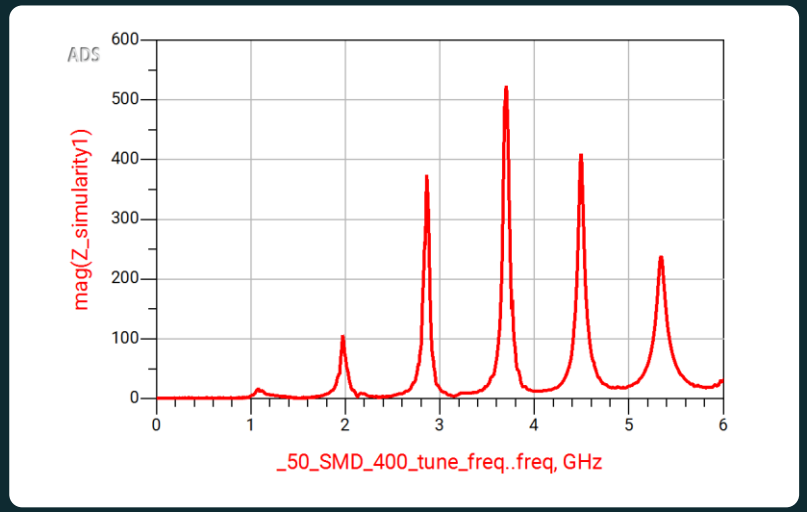
Above about 3 GHz the |Z| resonances shift and |ΔZ| exceeds 30 Ω: parasitics outside the model take over.

SMD 50 Ω, 400 mil: model holds up to about 1.8 GHz









Element	Fitted value
R	49.15 Ω
L	8.98 nH
C1 / C2	0.01 / 0.16 pF
SMA cap / ind	0.230 pF / 10 pH
Adapter / SMA delay	0.090 / 0.048 ns

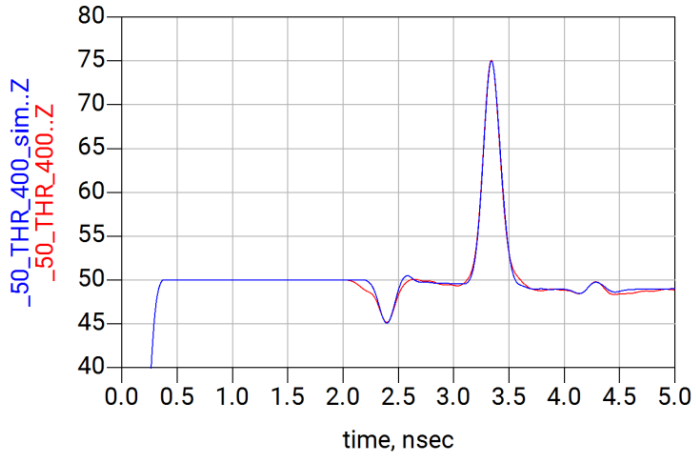
Key results

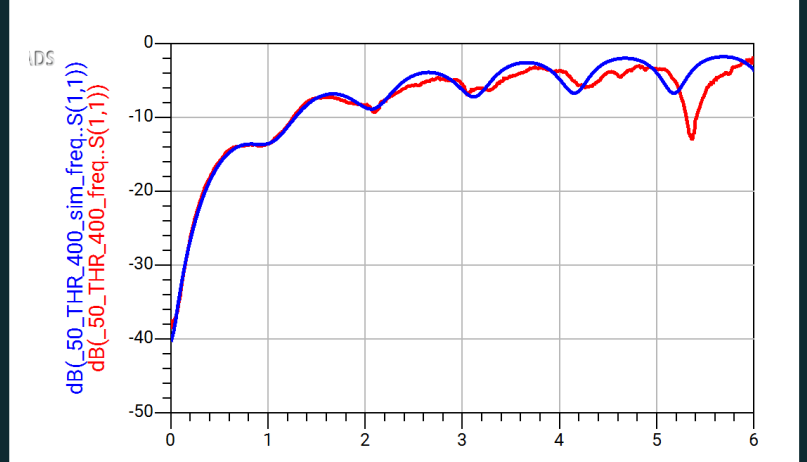
- TDR peak about 107 Ω, matched
- Resonance frequencies line up to about 3 GHz
- |ΔZ| under about 15 Ω to 1.8 GHz

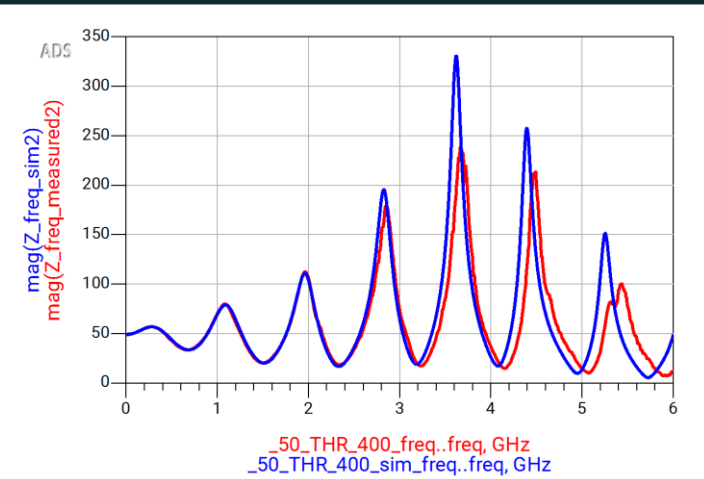
SO, WHAT

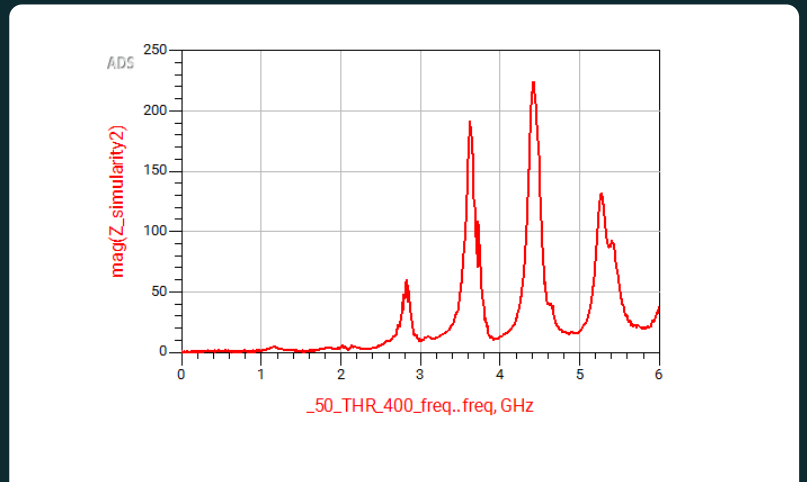
The 400 mil loop makes |Z| sharply resonant; small frequency offsets produce large |ΔZ| at each peak.

Through-hole 50 Ω, 400 mil: model holds up to about 2.5 GHz









Element	Fitted value
R	49.05 Ω
L	5.18 nH
C1 / C2	0.31 / 0.01 pF
SMA cap / ind	0.340 pF / 10 pH
Adapter / SMA delay	0.100 / 0.041 ns

Key results

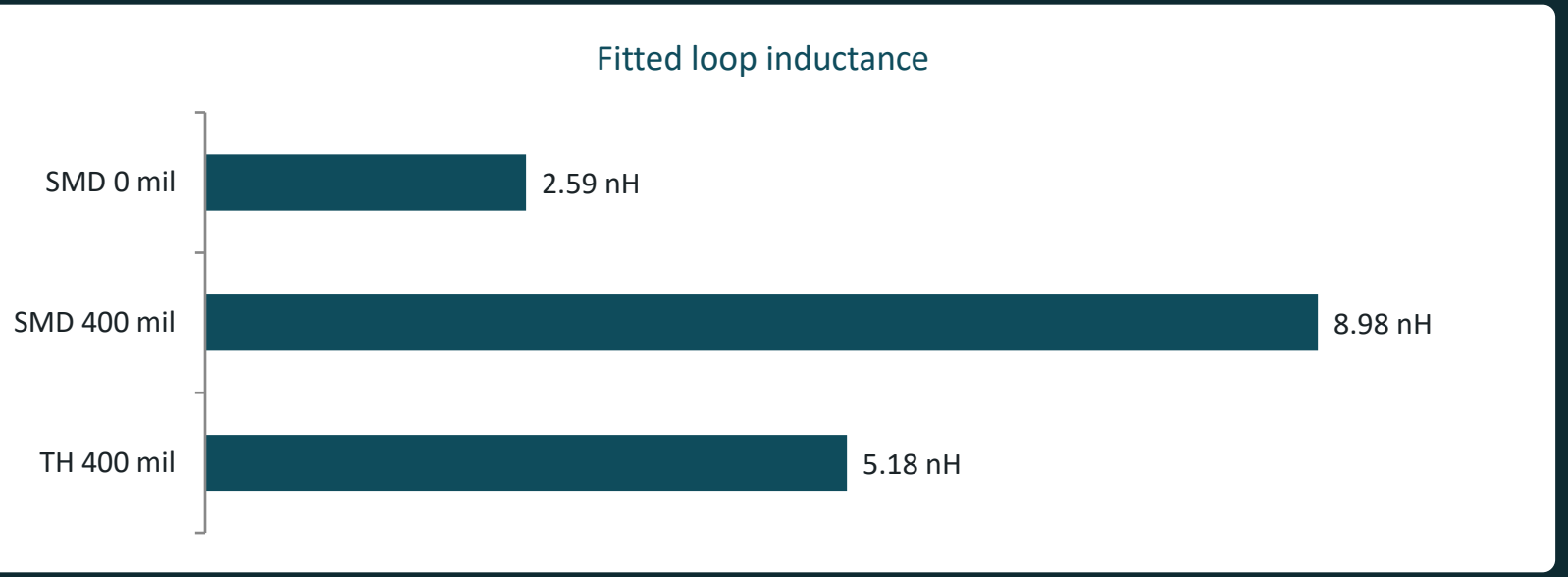
- TDR peak about 75 Ω, matched
- |S11| within about 1 dB to 3 GHz
- |ΔZ| under about 10 Ω to 2.5 GHz

SO, WHAT

Lower L than SMD at the same spacing (5.18 vs 8.98 nH), so a lower peak; above about 3 GHz the model resonances run ahead of the measurement.

Geometry sets inductance, and inductance sets bandwidth

Parameter	SMD 0 mil	SMD 400 mil	TH 400 mil
R (Ω)	48.95	49.15	49.05
L (nH)	2.59	8.98	5.18
C1 / C2 (pF)	0.39 / 0.01	0.01 / 0.16	0.31 / 0.01
TDR peak (Ω)	≈ 58	≈ 107	≈ 75
Model valid to (GHz)	≈ 2	≈ 1.8	≈ 2.5



SO, WHAT

Moving the via 400 mil away adds about 6.4 nH (2.59 → 8.98 nH), nearly doubles the TDR peak and cuts the usable bandwidth. Package and layout set the highest frequency at which a termination still looks like a resistor.

Questions?